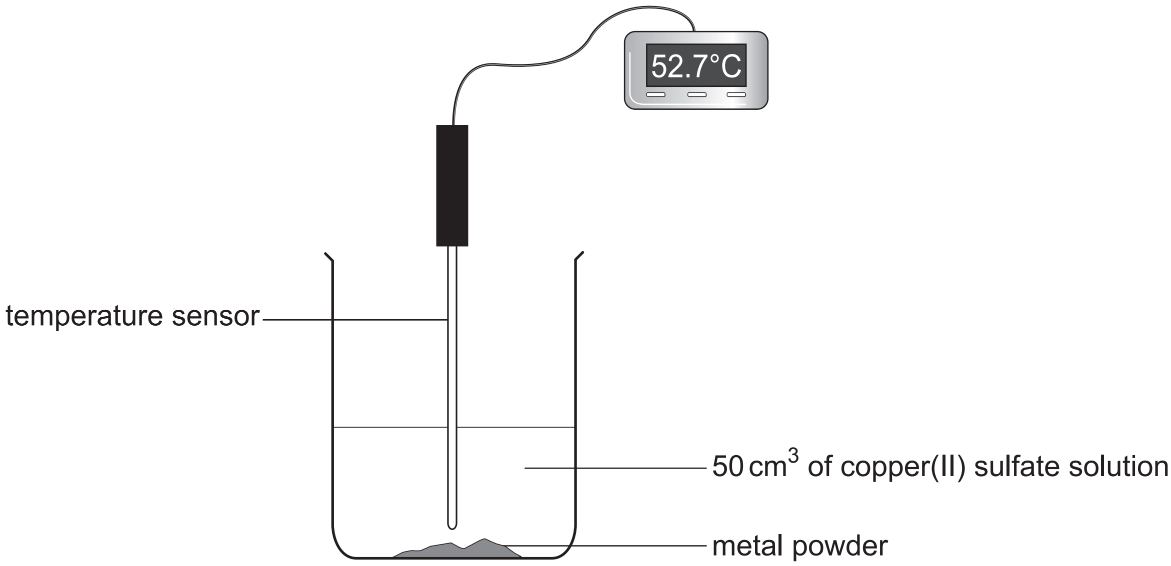
9. The list below shows part of the reactivity series.

magnesium
zinc
iron
nickel

A student investigated the temperature rise when four different metal powders were added to excess copper(II) sulfate solution.

The same mass of each metal was added to 50 cm^3 samples of copper(II) sulfate solution.

The maximum temperature for each reaction was measured. The temperature rise was calculated in each case and used to find the energy given out.



The results are shown in the table below.

Metal	Temperature rise (°C)	Energy given out (J)
magnesium	40.5	8500
zinc	33.0	6900
iron	23.2	4900
nickel	19.0	4000

Examiner only

(a) In one of the reactions, the initial temperature was 19.7 °C and the maximum temperature was 52.7 °C.

State which one of the four metals was used in this reaction. [1]

....

(b) Tick (✓) the box next to the conclusion the student can draw from the results. [1]

The higher the metal in the reactivity series, the greater the energy given out	☐
The lower the metal in the reactivity series, the greater the energy given out	☐
The energy given out is not related to the metal's position in the reactivity series	☐

(c) The word equation below represents the reaction between iron and nickel(II) sulfate solution.

iron + nickel(II) sulfate ⟶ iron(II) sulfate + nickel

Iron(II) sulfate contains Fe^{2+} and SO_4^{2-} ions.

Complete the symbol equation for the reaction. [2]

$Fe + NiSO_4 \longrightarrow$ .... + ...

(d) The experiment shows that a more reactive metal will replace a less reactive metal in its compounds.

Give the term used to describe this type of reaction. [1]

....

Examiner only

(e) When the experiment was repeated using **titanium**, the temperature rise recorded was 35.4 °C.

Calculate the energy given out during the reaction between **titanium** and 50 cm^3 of copper(II) sulfate solution. Give your answer to the nearest 100 J. [2]

$$\text{energy given out (J)} = \text{volume of solution} \times 4.2 \times \text{temperature rise}$$

Energy given out = ... J

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